Computer Systems

(CS15 – PC Assembly and Troubleshooting)

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COMPUTER SYSTEM

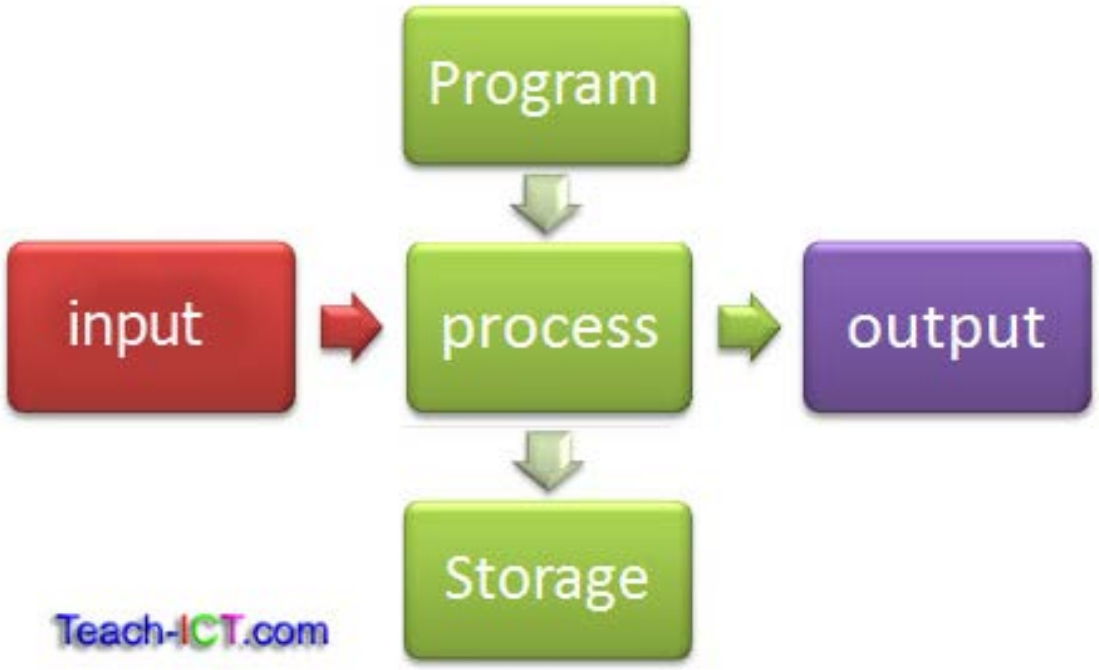
A complete set working computer.

Computer systems will include the computer along with any software and peripheral devices that are necessary to make the computer function.

In a small business, it's a system of interconnected computers that share a central storage system and various peripheral devices such as printers, scanners, Network Access Storage (NAS) Device, or Routers.

A computer system is one that is able to take a set of inputs, process them and create a set of outputs. This is done by a combination of hardware and software.

The diagram below shows you the idea of a computer system in its most basic form



The computer system has one or more ***inputs*** to provide data. This data is then ***processed*** in some way. The outcome of the processing is sent to an ***output*** or it may be ***stored*** until some event happens to cause it to be output.

For processing to take place, there needs to be a set of instructions of what needs to be done. This set of instructions is called a ***program***.

Computer systems may be classified into the following groups:

The ***Super Computers*** are extremely fast computers that are used in scientific research for their ability to do large amount of calculations and store large amount of data.

Sequoia Super Computer (U.S.)

More SuperComputers @ http://www.peimag.com/top-10-most-powerful-supercomputers-in-the-world/

Mainframe Computers allows several thousands of users at any time. They are used by government companies, airline companies for bookings, insurance companies and banks.

a technician preparing IBM zEnterprise EC12 Mainframe computers for shipment

Personal Computers are for personal use at home or at work. In what follows, the different types of personal computers are ….

Desktop

Laptop

Netbook / Tablet

Smartphones

http://www.newtechnologysite.com/computers/types.html

Desktop

It is a computer system that is used at home, at work, in schools, universities... It consists of a monitor and a casing inside which the power supply, the CPU, the motherboard, hard disk, and many other devices are located.

These computers may be used for creating, editing, printing and sending documents. They are also used to communicate using the vast network of computers spread all over the world called the internet. They are used to play music, videos, games, and learn new languages, mathematics and many other scientific subjects.

Laptop has similar computing capabilities as the desktop but has a set of electrical batteries that allows it to be used in cars, parks, airplanes. They also have a smaller weight and are therefore portable. Besides being used to surf the net, send e-mails, they can also be used to create, edit and send documents, excel sheets. The portability and the computing power of laptops make them suitable for presentations, using for example power point, in conferences and forums, they also have enough power to play games, read and write CD's and DVD's.

Netbook specializes in browsing the Web. They have the looks of laptops, but lack the full capabilities of a computer. They weigh less than a laptop and are smaller in size which are big advantages for travelers with needs to these machines. Almost all netbooks do not have CD / DVD drives. They also have smaller screens and their keyboards are smaller too.

A ***Tablet PC*** has the capabilities of a laptop but is smaller in size and does not have a keyboard. It has a touch-screen that allows using fingers or stylus to input instructions and interact with the machine. The keyboard or mouse are not needed. It is light and small in size and is therefore easy to transport.

Modern Tablet PCs allow you to use the internet to look for information and news, watch online videos, read e-books, communicate with your friends, shop online. Tablets are fitted with batteries that may last up to 10 hours. Major advances in technologies such as battery life, smaller size yet larger memories, touch-screen recognition, and wireless internet access have all contributed to make the use of tablets PC more efficient and practical.

Besides making telephone calls, a ***Smartphone*** runs on an operating system (Android / IOS) and has the ability to let you send and receive e-mails, browse websites on the net and read documents.

Computer Capabilities

- Ability to Perform Certain Logic Operations
- Ability to Provide New Time Dimensions
- Ability to Store (Save) and Retrieve (Re-Open) Data / Information
- Ability to Control Error
- Ability to Check Itself

Referenced from the original ComSci black book:
INTRODUCTION TO COMPUTER CONCEPTS
by Juny Pilapil La Putt

Computer Limitations

- Dependence on Prepared Instructions
- Inability to Derive Meanings from Objects
- Inability to Generate Information
- It cannot correct certain wrong inputs / instruction

Referenced from the original ComSci black book:
INTRODUCTION TO COMPUTER CONCEPTS
by Juny Pilapil La Putt

Why Computer Sometimes Fail

- Input Errors
- Errors in instructing / programming a computer
- Communication Gap
- Improper Controls / Use
- Lack of Standards
- Lack of Adequate Manufacturer Support

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